MODEL QUESTION PAPER 2

S4 INSTRUMENTATION ENGINEERING

SUBJECT: ANALYTICAL & BIOMEDICAL INSTRUMENTS

Time: 3 Hour *Max. Marks: 75*

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		Module Outcomes	*Cognitive Level*
1	List the basic components of a spectrometer.	M1.08	R
2	Name any two UV sources.	M1.04	R
3	Name any two carrier gases used in gas chromatography.	M2.03	U
4	What is HPLC?	M2.05	R
5	Define chromatography.	M2.01	R
6	Define the term thermal conductivity of a gas.	M3.02	R
7	What do you mean by the paramagnetic nature of oxygen?	M3.03	U
8	What are the typical values of action and resting potential of a cell?	M4.02	R
9	Draw an Einthoven triangle.	M4.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		Module Outcomes	*Cognitive Level*
1.	State Beer Lambert's law.	M1.02	R
2.	Explain the working of thermal conductivity type gas analyzer.	M3.02	U
3.	Explain the various regions of electromagnetic spectrum.	M1.01	R
4.	What is the basic principle behind liquid chromatography? What are its types?	M2.05	U
5.	Write a note on paper chromatography.	M2.03	U
6.	Explain the construction of zirconia oxygen analyzer	M3.04	U
7.	Explain briefly about positive filter type IR analyzer.	M3.05	U
8.	What are the various frequency regions of EEG spectrum?	M4.06	R
9.	Draw an ECG waveform and explain what each region represents.	M4.03	U
10.	Explain any one detector used in HPLC.	M2.05	U

Part – C

Answer all questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		Module Outcomes	*Cognitive Level*
III	Define Raman Effect. With neat block diagram, explain Raman spectrometer.	M1.10	U
	OR		
IV	IExplain the construction and working of magnetic deflection type mass spectrometer.	M1.08	U
V	Discuss the basic principle behind flame photometer and explain its construction.	M1.06	U
	OR		
VI	Explain the working of dual beam filter photometer with a neat sketch.	M1.05	U
VII	With a neat diagram, explain the working of gas chromatography.	M2.04	U
	OR		
VIII	Explain any two detectors used in gas chromatography.	M2.04	U
IX	Describe the construction and working of a magnetic wind type oxygen analyzer.	M3.03	U
	OR		
X	Define electrical conductivity. Discuss any one method to measure electrical conductivity.	M3.06	U
XI	Explain the block diagram of an ECG machine.	M4.04	U
	OR		
XII	Describe the electrodes used for ECG measurement.	M4.03	U
XIII	Explain in detail about 10-20 electrode lead system of EEG	M4.06	U
	OR		
XIV	What is EMG? Describe the block diagram of an EMG machine.	M4.05	U

Question Wise Analysis

Course: **ANALYTICAL & BIOMEDICAL INSTRUMENTS** – Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.08	R	1	2
I.2	M1.04	R	1	2
I.3	M2.03	U	1	2
I.4	M2.05	R	1	2
I.5	M2.01	R	1	2
I.6	M3.02	R	1	2
I.7	M3.03	U	1	2
I.8	M4.02	R	1	2
I.9	M4.03	R	1	2
II.1	M1.02	R	3	8
II.2	M3.02	U	3	8
II.3	M1.01	R	3	8
II.4	M2.05	U	3	8
II.5	M2.03	U	3	8
II.6	M3.04	U	3	8
II.7	M3.05	U	3	8
II.8	M4.06	R	3	8
II.9	M4.03	U	3	8
II.10	M2.05	U	3	8
III	M1.10	U	7	16
IV	M1.08	U	7	16
V	M1.06	U	7	16
VI	M1.05	U	7	16
VII	M2.04	U	7	16
VIII	M2.04	U	7	16
IX	M3.03	U	7	16
X	M3.06	U	7	16
XI	M4.04	U	7	16
XII	M4.03	U	7	16
XIII	M4.06	U	7	16
XIV	M4.05	U	7	16

Course : **ANALYTICAL & BIOMEDICAL INSTRUMENTS –QP 2**

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module $(h_i / \sum H_i) * 123\ (\pm 5\%)$	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	11	30 ± 6	2	2	2	6	4	28	8	36
2	11	30 ± 6	3	3	3	9	2	14	8	26
3	11	30 ± 6	2	2	3	9	2	14	7	25
4	12	33 ± 6	2	2	2	6	4	28	8	36
Total	45	123	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	16	13
Understanding	107	87
Applying	0	0
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100